Distributed gradient descent: splitting the batch

Part 1 of 2: when the model fits and the data does not

Carlos Cotrini

Section 1

The gradient of a mean

1

The gradient of a mean

Paris, 1793: one task, twenty computers

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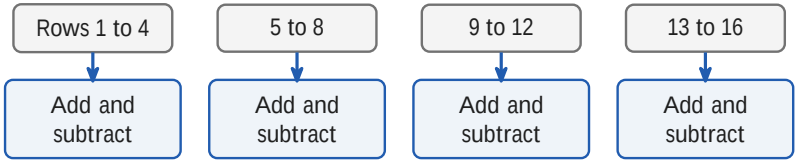
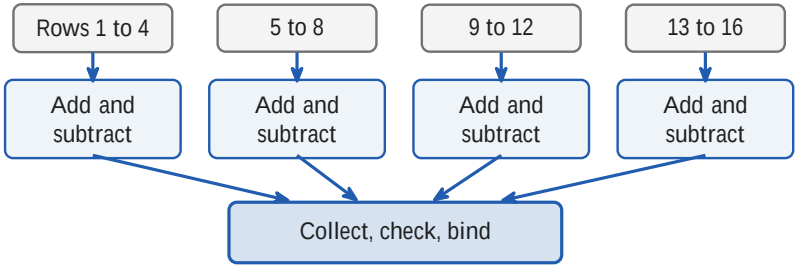


Paris, 1793: one task, twenty computers



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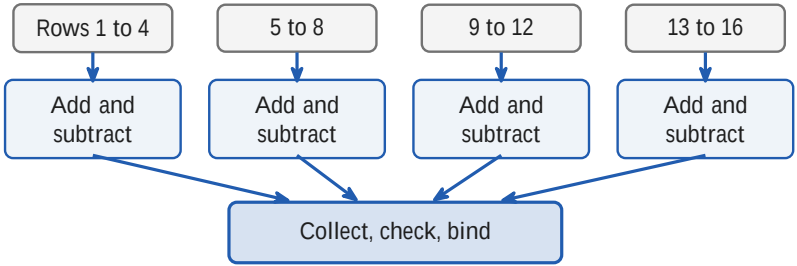

- The same arithmetic in every hand, a di□erent slice of the table

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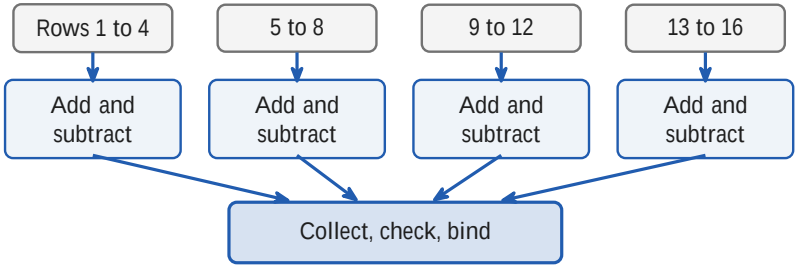
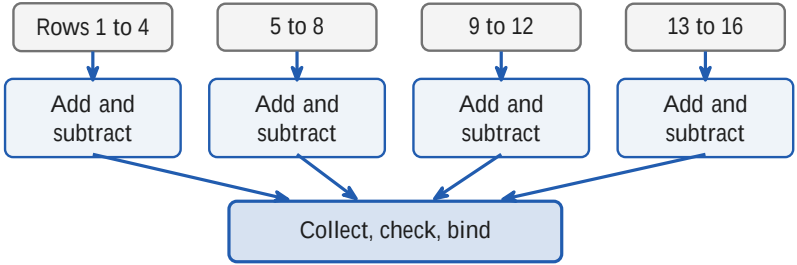


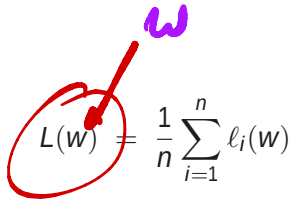
- The same arithmetic in every hand, a di□erent slice of the table
- Not a pin factory: nobody specialises in a step

Paris, 1793: one task, twenty computers



- The same arithmetic in every hand, a di□erent slice of the table
- Not a pin factory: nobody specialises in a step
- Roegel 2010, from the payrolls: about twenty computers, not the sixty to eighty Prony recalled.

The sum splits, exactly



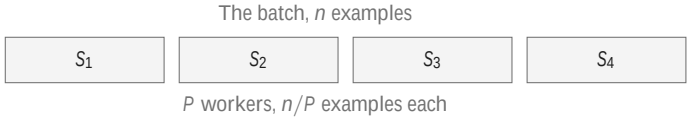
The sum splits, exactly

$$\nabla L(w) = \frac{1}{n}\sum_{i=1}^{n} \nabla \ell_i(w)$$

$$\nabla L(w) = \frac{\nabla \ell_1(w) + \nabla \ell_2(w) + \nabla \ell_3(w) + \dots + \nabla \ell_n(w)}{n}$$

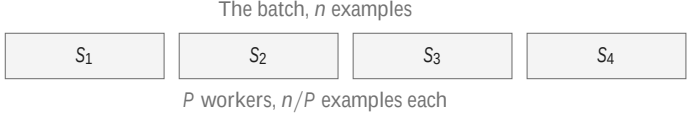
The sum splits, exactly

$$\nabla L(w) \;=\; \frac{1}{P}\sum_{p=1}^{P} \underbrace{\frac{P}{n}\sum_{i\in S_p}\nabla \ell_i(w)}_{\text{worker } p \text{ computes this alone}}$$



The sum splits, exactly

$$\nabla L(w) \;=\; \underbrace{\frac{1}{P}\sum_{p=1}^{P} g_p}_{\text{the only thing the workers must share}}$$



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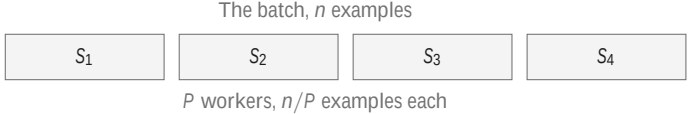

- Equal shards, so every worker holds the same number of examples

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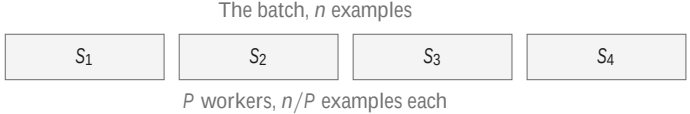
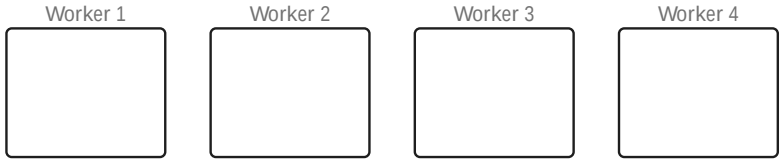


- Equal shards, so every worker holds the same number of examples
- You already have this from Weekend 0. Nothing here is new mathematics.

Every worker holds the whole model



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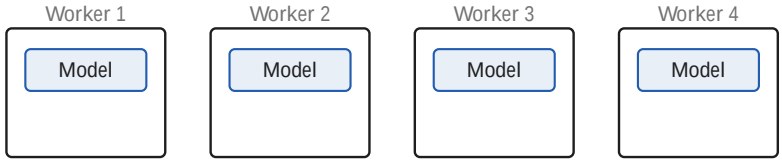

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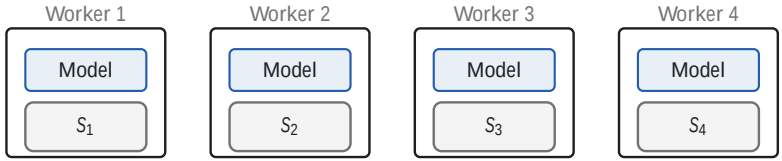

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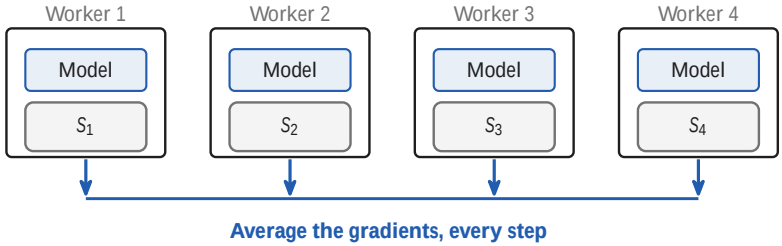

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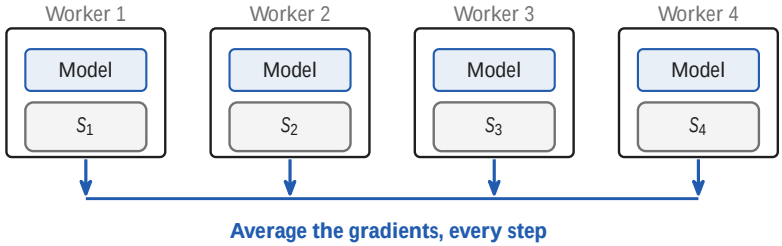

- Data parallelism: replicate the model, split the batch

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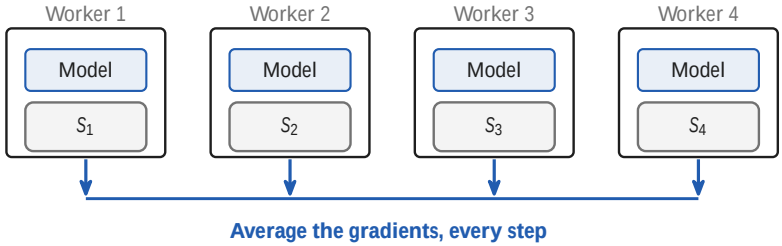

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- The copies stay identical because every worker applies the same average

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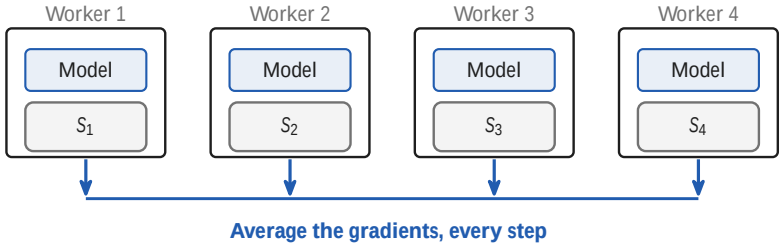
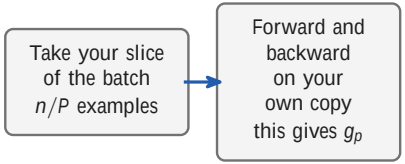


- Data parallelism: replicate the model, split the batch
- The copies stay identical because every worker applies the same average
- This hour assumes the model fits on one device

One step of training, run on every worker at once

Take your slice of the batch n/P examples

One step of training, run on every worker at once



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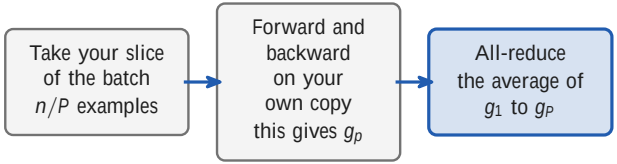

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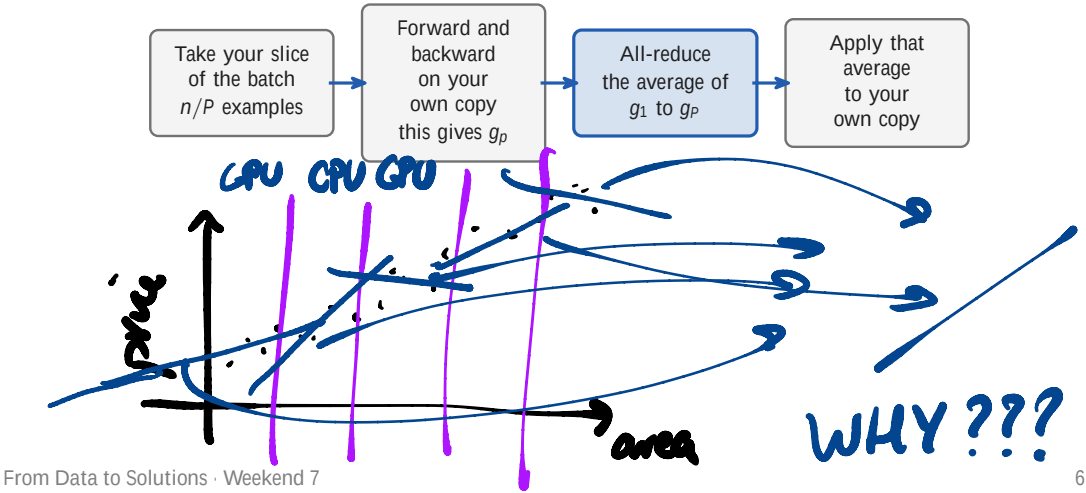

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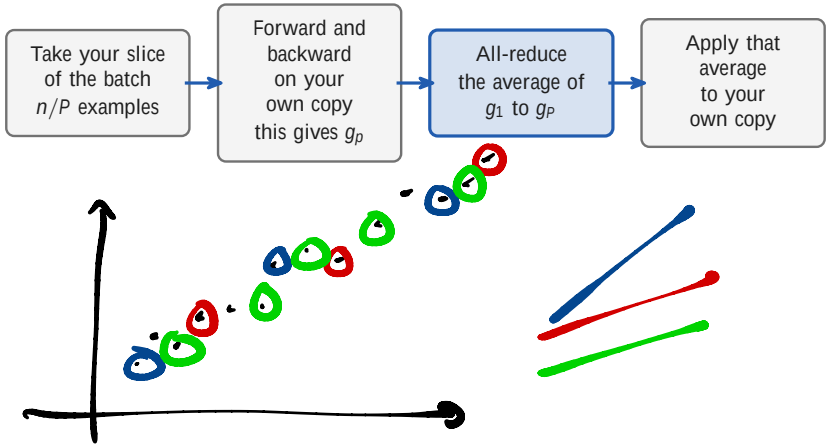

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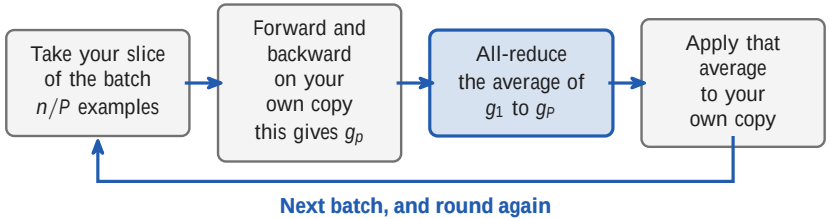

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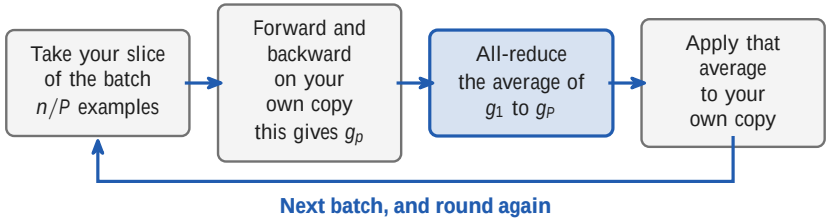
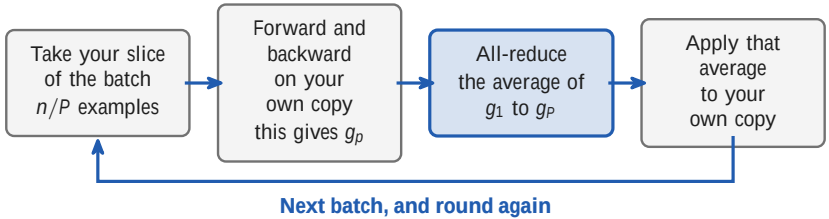


Three of the four stages are local. The all-reduce is the one that touches the network.

One step of training, run on every worker at once



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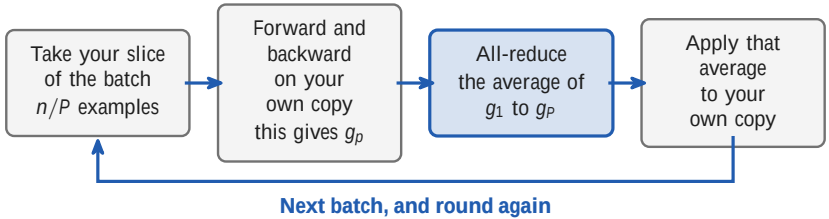


Three of the four stages are local. The all-reduce is the one that touches the network.

- Every worker runs this same loop, at the same time, on its own slice
- Split the data, aggregate the gradient, and repeat

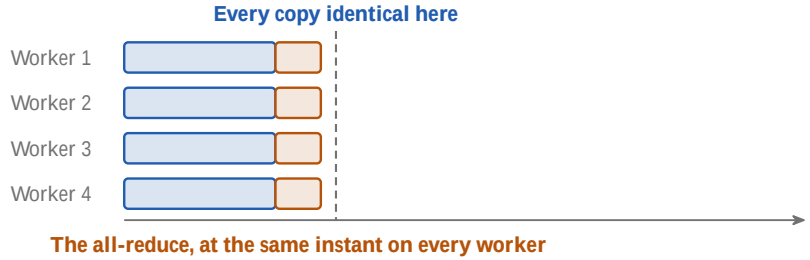
One all-reduce per step, for every step of the run

The gradient of a mean

One all-reduce per step, for every step of the run

The gradient of a mean

One all-reduce per step, for every step of the run



The gradient of a mean

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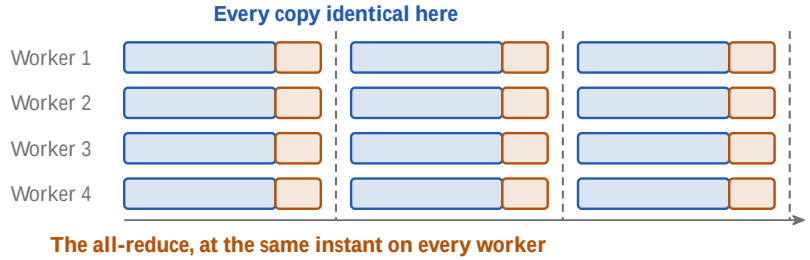

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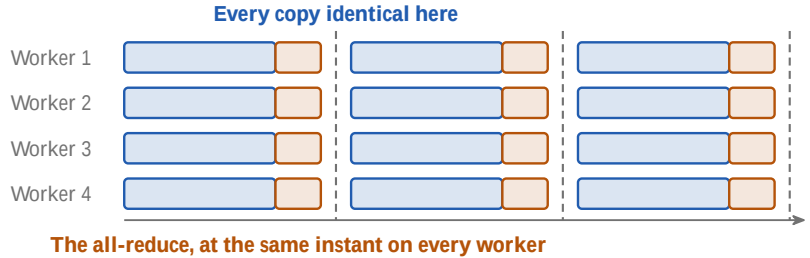
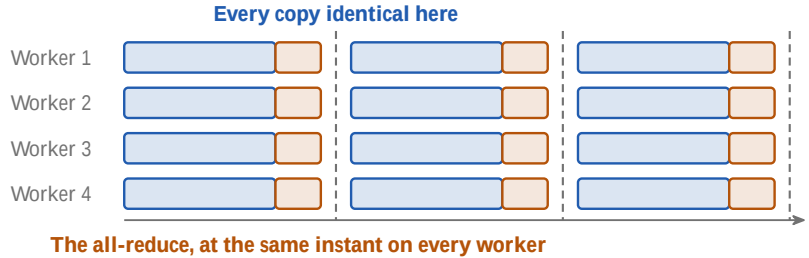


- The average lands before the weights move, so every step starts from one model

The gradient of a mean

One all-reduce per step, for every step of the run



- The average lands before the weights move, so every step starts from one model
- Local SGD relaxes this and aggregates every few steps. Everything in this hour is the synchronous version.

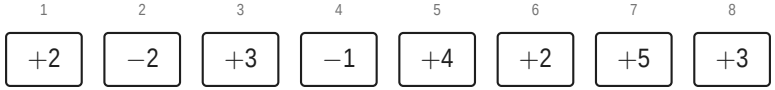
Section 2

Four workers, eight examples

2

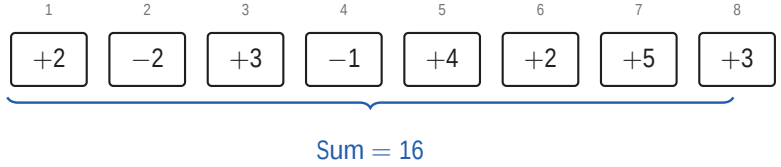
Four workers, eight examples

Eight examples, one gradient



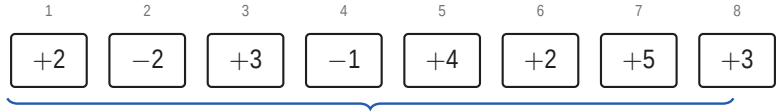
Four workers, eight examples

Eight examples, one gradient



Four workers, eight examples

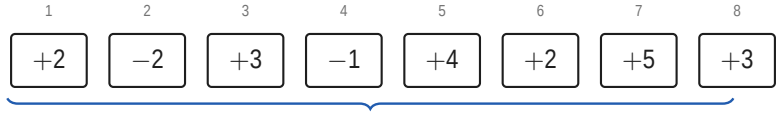
Eight examples, one gradient



- The single machine answer is 2.0

Four workers, eight examples

Eight examples, one gradient

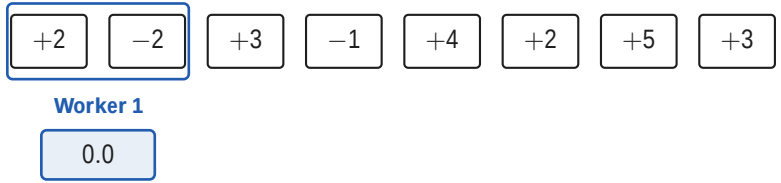


- The single machine answer is 2.0
- Constructed for this lecture. A real gradient is a vector with one number per parameter; one number per example keeps the arithmetic hand checkable.

Four workers, eight examples

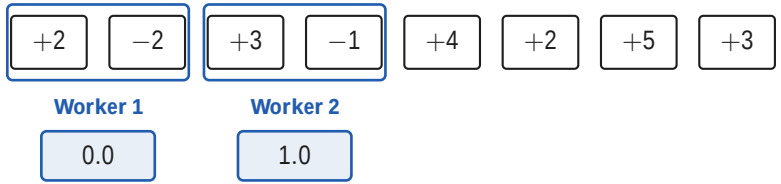
Four workers, two examples each

Four workers, two examples each



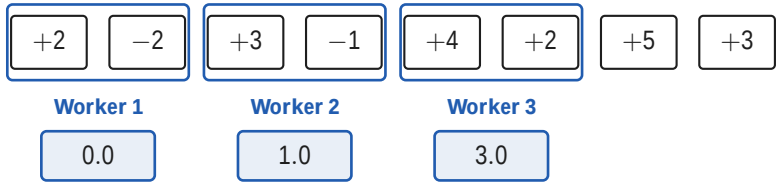
Four workers, eight examples

Four workers, two examples each



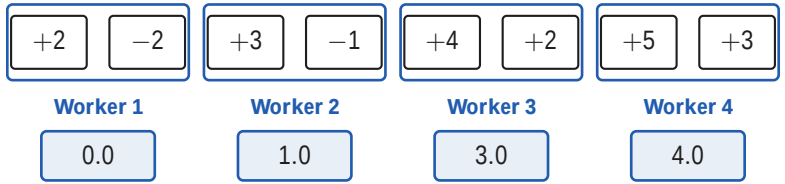
Four workers, eight examples

Four workers, two examples each

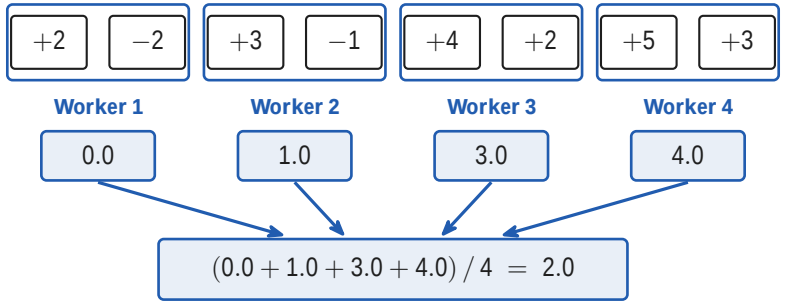


Four workers, eight examples

Four workers, two examples each



Four workers, two examples each



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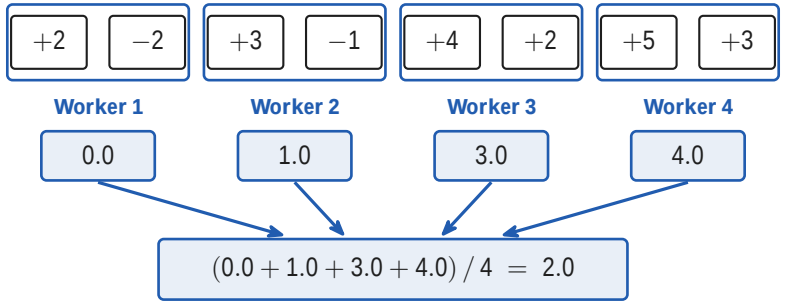
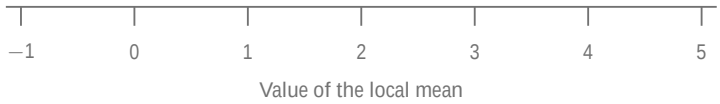


Exactly equal, not approximately: splitting the batch is a regrouping of the same sum.

Four workers, eight examples

Worker 1 sees zero



Four workers, eight examples

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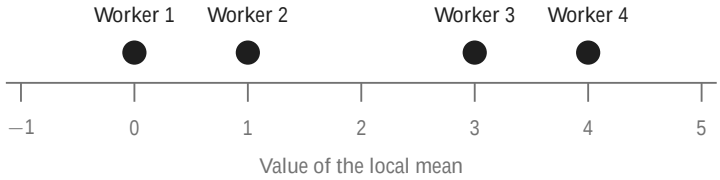

Four workers, eight examples

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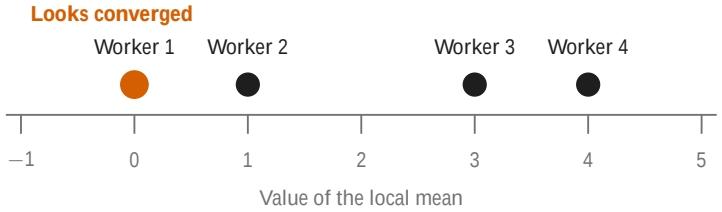
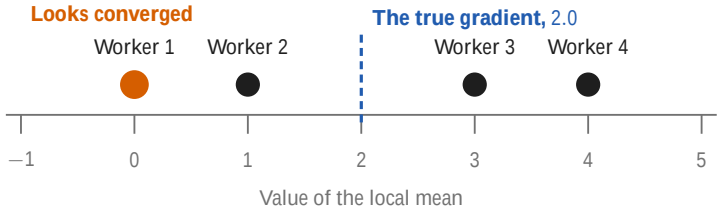


- Worker 1’s local mean is zero. Its own shard looks converged.

Four workers, eight examples

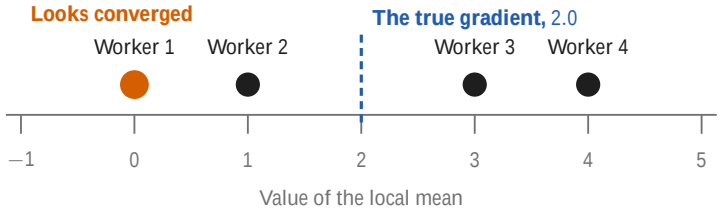
Worker 1 sees zero



- Worker 1's local mean is zero. Its own shard looks converged.
- The global answer is 2.0, and no worker holds it

Four workers, eight examples

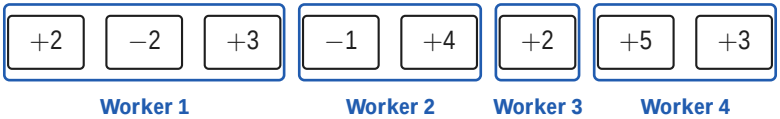
Worker 1 sees zero



- Worker 1's local mean is zero. Its own shard looks converged.
- The global answer is 2.0, and no worker holds it
- That is why the averaging step exists, and why it cannot be skipped

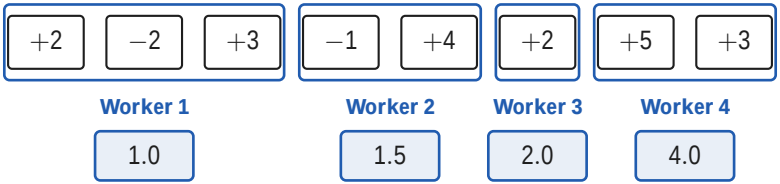
Four workers, eight examples

Unequal shards break the identity



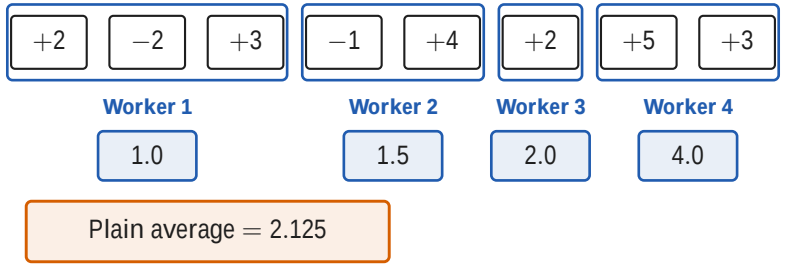
Four workers, eight examples

Unequal shards break the identity



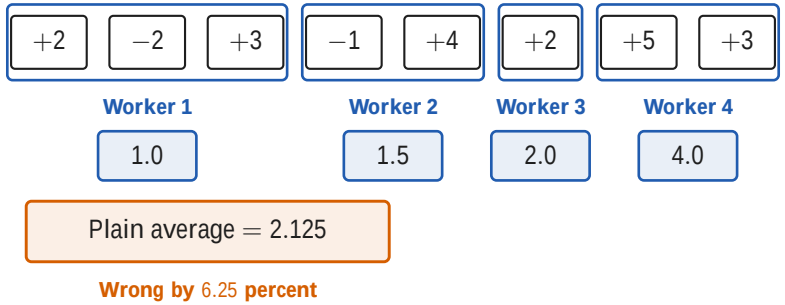
Four workers, eight examples

Unequal shards break the identity



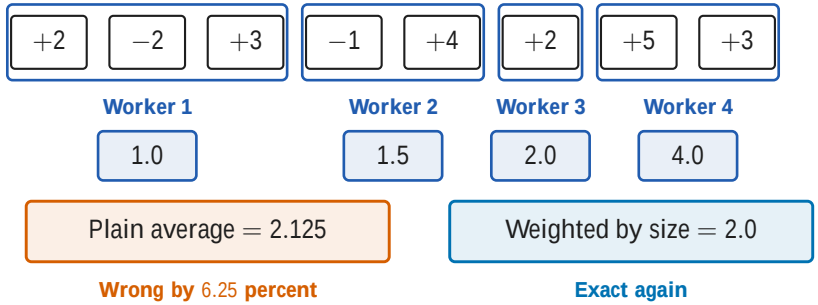
Four workers, eight examples

Unequal shards break the identity



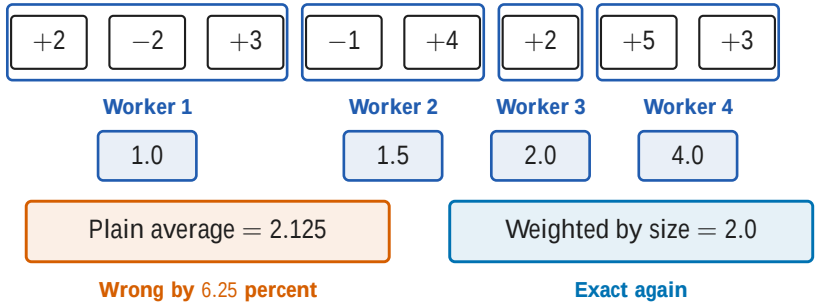
Four workers, eight examples

Unequal shards break the identity



Four workers, eight examples

Unequal shards break the identity



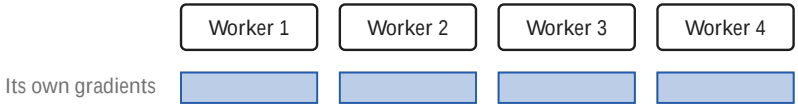
Every framework therefore fixes the per worker micro batch and drops or pads the ragged tail.

Four workers, eight examples

One bu□er, 16.11 gigabytes

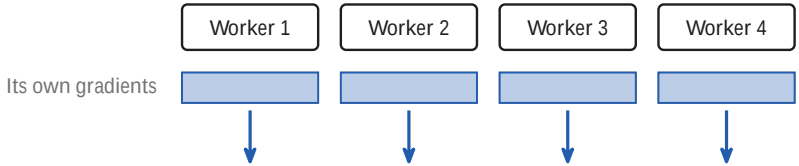
Four workers, eight examples

One bu□er, 16.11 gigabytes



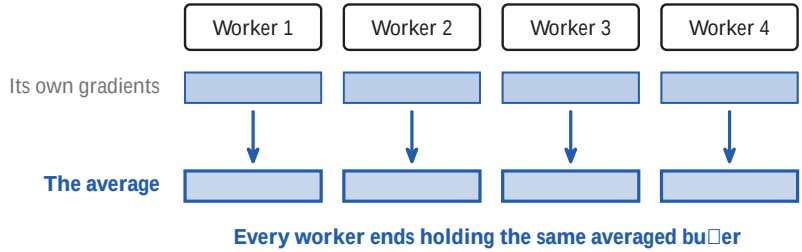
Four workers, eight examples

One bu□er, 16.11 gigabytes



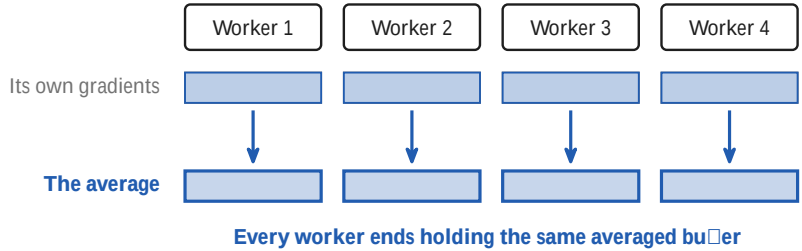
Four workers, eight examples

One bu□er, 16.11 gigabytes



Four workers, eight examples

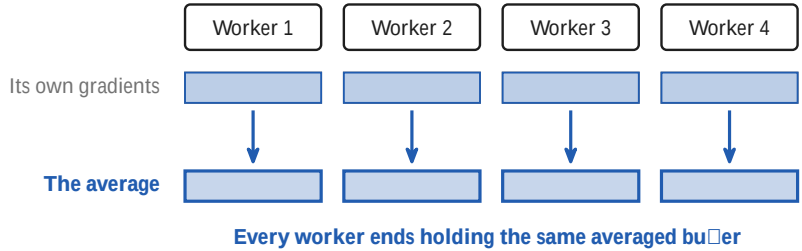
One bu□er, 16.11 gigabytes



- Apertus-8B in bf16: 2 bytes $\times$ 8.053 $\times$ 10^9 parameters $=$ 16.11 GB

Four workers, eight examples

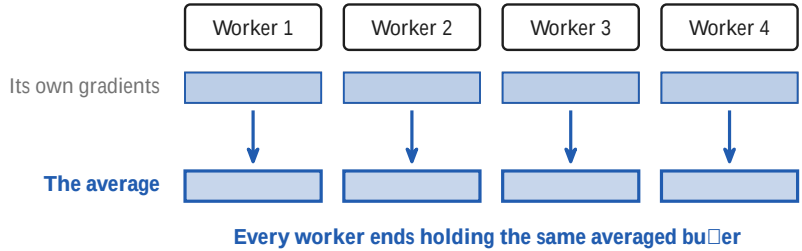
One bu□er, 16.11 gigabytes



- Apertus-8B in bf16: 2 bytes $\times$ 8.053 $\times$ 10^9 parameters $=$ 16.11 GB
- Write K for the size of that bu□er, and P for the number of workers

Four workers, eight examples

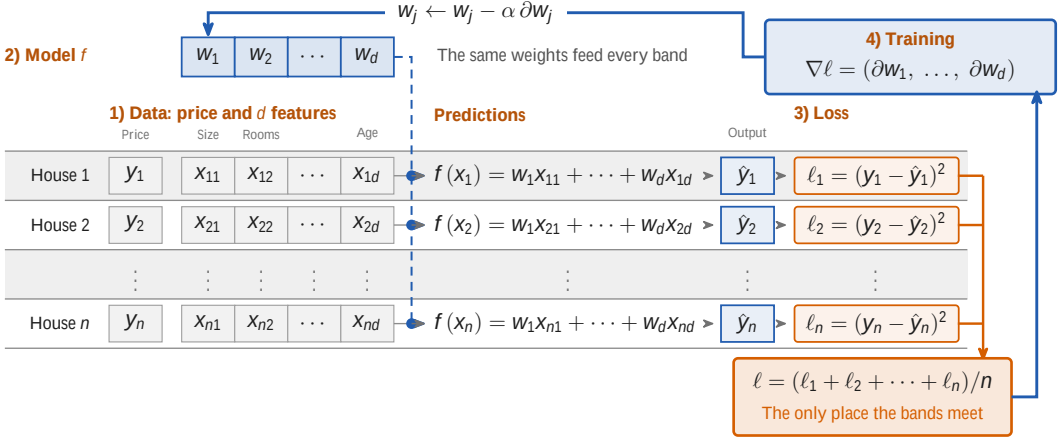
One bu□er, 16.11 gigabytes



- Apertus-8B in bf16: 2 bytes $\times$ 8.053 $\times$ 10^9 parameters $=$ 16.11 GB
- Write K for the size of that bu□er, and P for the number of workers
- Compute is local. This is the only part that is not.

Four workers, eight examples

One training step of linear regression

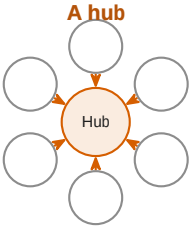


Section 3

The all-reduce

3

Two ways to average: a hub, or a ring

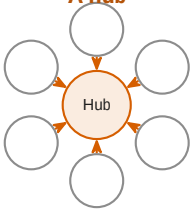


K is the gradient bu□er, 16.11 GB. *P* is the number of workers.

Two ways to average: a hub, or a ring

16 GB
↓
every ∇ weigths K Bytes

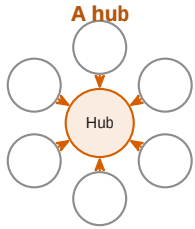
Total 6×16GB

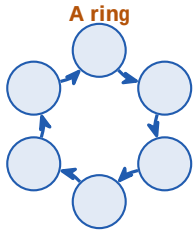


$P \times K$ **arrives at one machine**

K is the gradient bu□er, 16.11 GB. P is the number of workers.

Two ways to average: a hub, or a ring

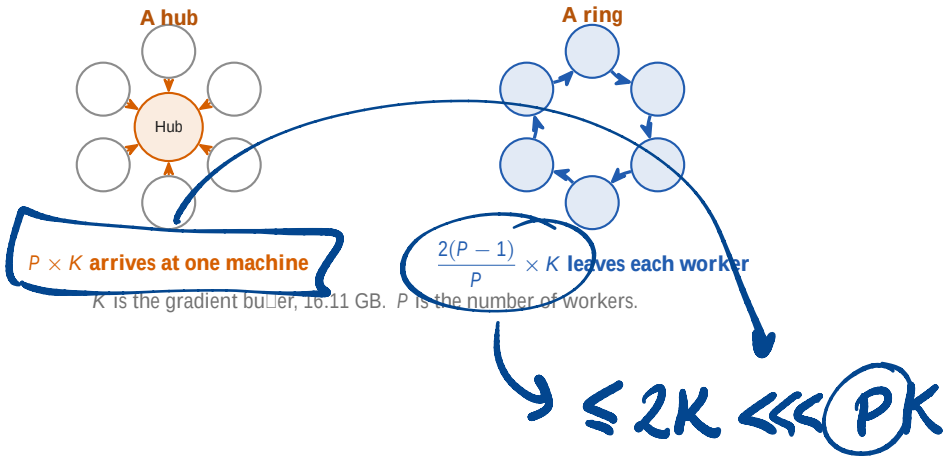




$P \times K$ **arrives at one machine**

K is the gradient bu□er, 16.11 GB. P is the number of workers.

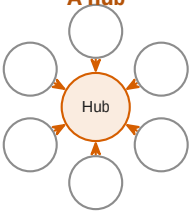
Two ways to average: a hub, or a ring



K is the gradient bu□er, 16.11 GB. P is the number of workers.

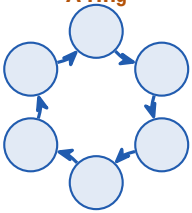
Two ways to average: a hub, or a ring

A hub



$P \times K$ **arrives at one machine**

A ring

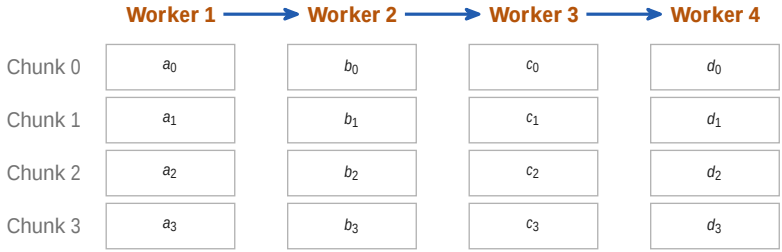


$\frac{2(P-1)}{P} \times K$ **leaves each worker**

K is the gradient bu□er, 16.11 GB. P is the number of workers.

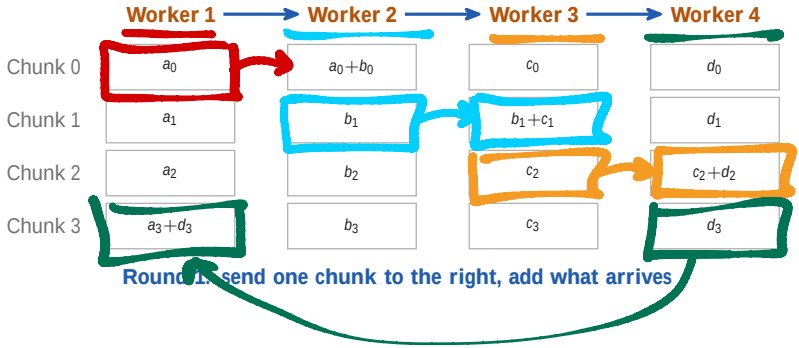
- The hub's tra□c grows with every worker you add

Round the ring, adding

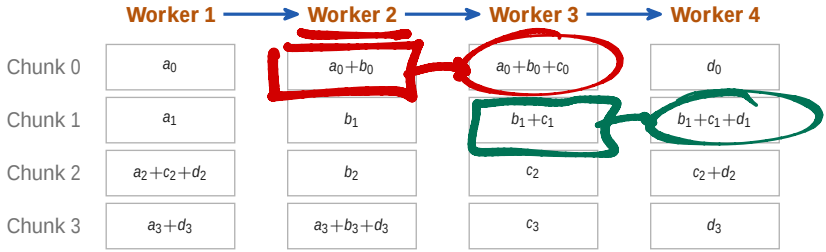


Start: every worker holds its own four chunks

Round the ring, adding

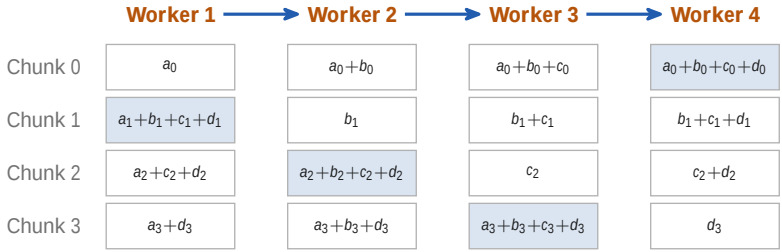


Round the ring, adding



Round 2: send on the chunk you have just added to

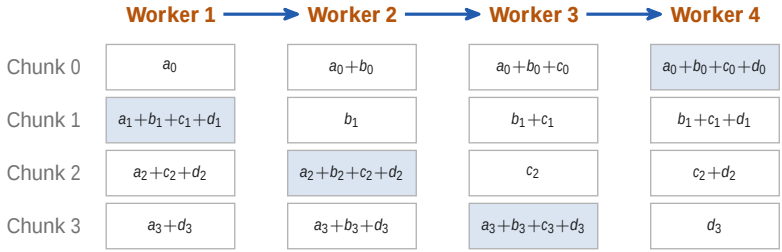
Round the ring, adding



Round 3: each worker now owns one chunk of the true sum

Round the ring, adding

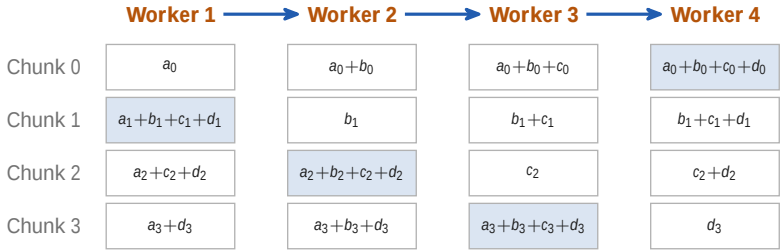
(P-1)



Round 3: each worker now owns one chunk of the true sum

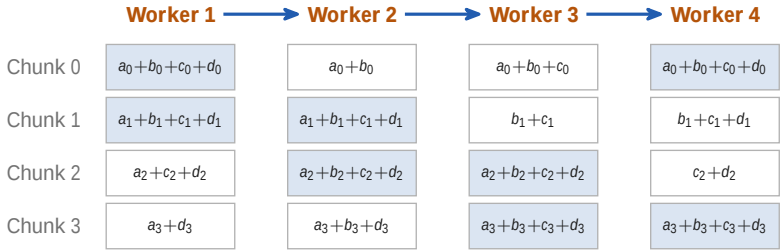
Chunk j is the same slice of the bu□er on every worker, so a_j is worker 1's chunk j, b_j worker 2's, and worker 4 sends back to worker 1.

Round the ring again, copying



Where the first half ended: one complete chunk each

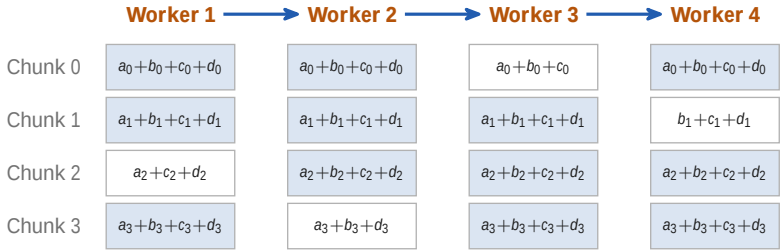
Round the ring again, copying



Round 4: send the complete chunk on, and overwrite

The all-reduce

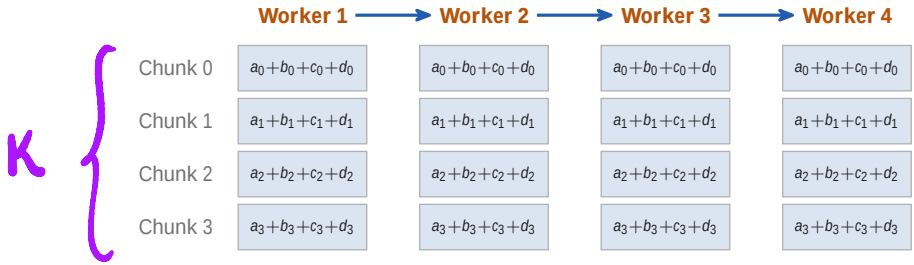
Round the ring again, copying



Round 5: the same tra□c, one hop further

Round the ring again, copying

P-1

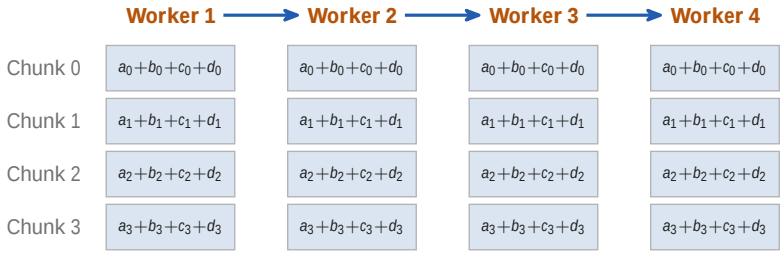


At each round

$$2\times(P-1)\frac{K}{P}$$

Round the ring again, copying

P-1



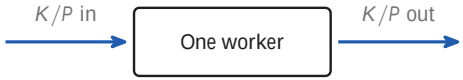
Round 6: every worker now holds the whole averaged bu□er

Every worker holds the whole model, so the copies stay identical only if every worker applies the same whole average

Same ring, same tra□c, one change: the arriving chunk replaces instead of adding. The four rows are four di□erent slices, so a column of four cells is the whole averaged bu□er.

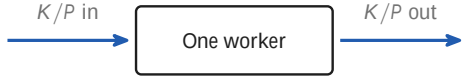
Two times P minus one rounds

$$\text{bytes per worker} \;=\; 2(P-1) \times \frac{K}{P}$$



Two times P minus one rounds

$$\text{bytes per worker} = \underbrace{2(P-1)}_{\text{rounds}} \times \underbrace{\frac{K}{P}}_{\text{one chunk}}$$

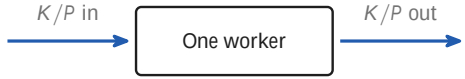


$P-1$ rounds adding, then $P-1$ copying

A chunk has to reach the other $P-1$ workers, so one pass round the ring is $P-1$ rounds, three of them at $P=4$

Two times P minus one rounds

$$\text{bytes per worker} = \underbrace{\frac{2(P-1)}{P}}_{\text{always below 2}} \times \underbrace{K}_{\text{the whole bu□er}}$$



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Two times P minus one rounds

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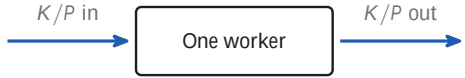


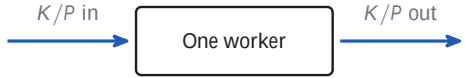
$P-1$ rounds adding, then $P-1$ copying

A chunk has to reach the other $P-1$ workers, so one pass round the ring is $P-1$ rounds, three of them at $P=4$

- The factor never reaches 2, so the tra□c never reaches $2K$

Two times P minus one rounds

$$\text{bytes per worker} = \underbrace{\frac{2(P-1)}{P}}_{\text{always below 2}} \times \underbrace{K}_{\text{the whole bu□er}}$$



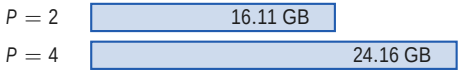
$P-1$ rounds adding, then $P-1$ copying

A chunk has to reach the other $P-1$ workers, so one pass round the ring is $P-1$ rounds, three of them at $P=4$

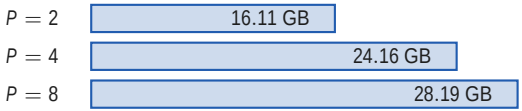
- The factor never reaches 2, so the tra□c never reaches $2K$
- Patarasuk and Yuan 2009 proved this is the least any all-reduce can move.

The bytes barely grow

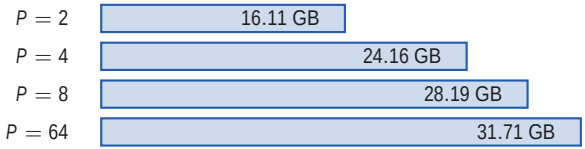
The bytes barely grow



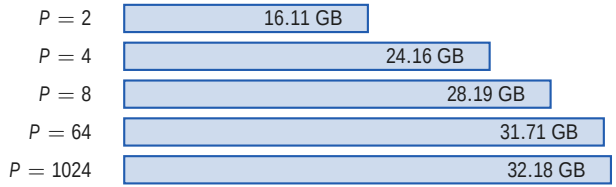
The bytes barely grow



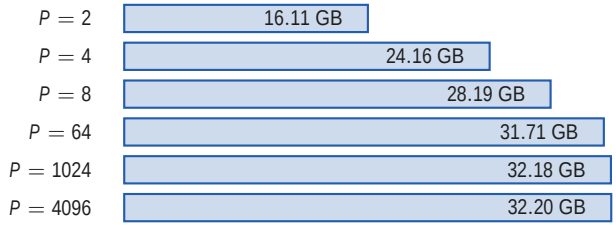
The bytes barely grow



The bytes barely grow



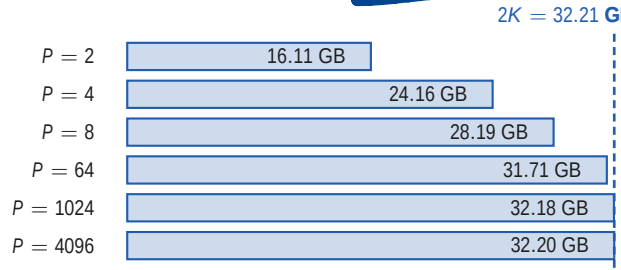
The bytes barely grow



The all-reduce

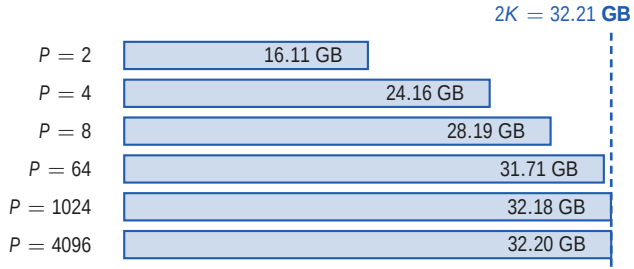
The bytes barely grow

$$\frac{2(P-1)}{P} \times K$$



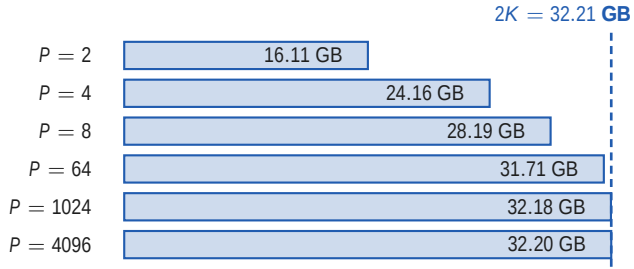
$$2 \cdot \frac{P-1}{P} \times K \qquad \leq 1 \qquad \leq 2K$$

The bytes barely grow



- 512 times the workers, 14.3 percent more bytes each

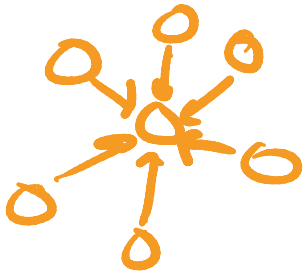
The bytes barely grow



- 512 times the workers, 14.3 percent more bytes each
- Apertus-8B gradients in bf16, so $K = 16.11$ GB.

The rounds grow linearly in P

$P = 8$ | 14 rounds



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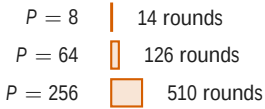

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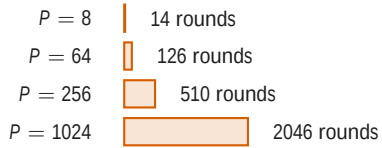

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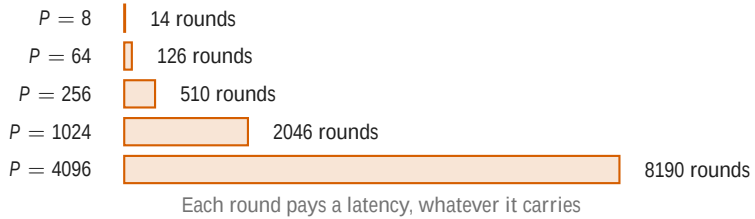

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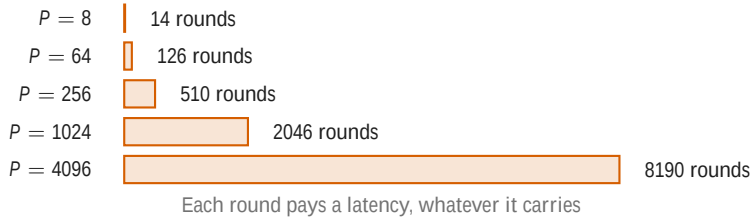

- The bytes barely grow. The rounds grow linearly.

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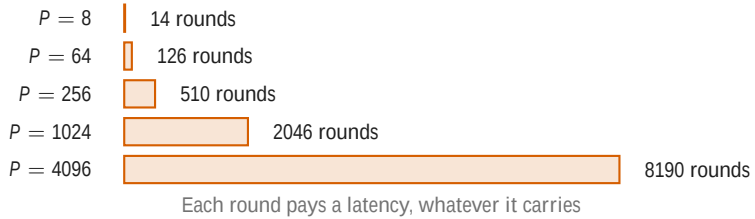

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- Past 512 GPUs the latency no longer hides behind the backward pass

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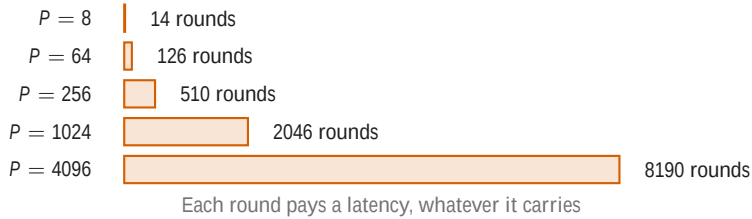

- The bytes barely grow. The rounds grow linearly.
- Past 512 GPUs the latency no longer hides behind the backward pass
- The ring is not Baidu's invention: it is Patarasuk and Yuan 2009.

Section 4

Where it stops scaling

4

One step, two terms

$T(P)$ is the wall clock time for *one training step* on P workers

$$T(P) = \frac{T(1)}{P} + \dots$$

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Where it stops scaling

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speed of transfer in B/s

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$$T(1024) = \underbrace{1128/1024}_{\text{compute}} + \underbrace{1.29}_{\text{all-reduce}} = 2.39 \text{ seconds}$$

Where it stops scaling

93 percent at 64 workers, 18 at 4,096

$$E(P) = \frac{T(1)}{P \times T(P)}$$

GPU-seconds of work, over GPU-seconds paid for

Workers	1	4	16
E□ciency, percent	**100**	**99.7**	**98.3**

Where it stops scaling

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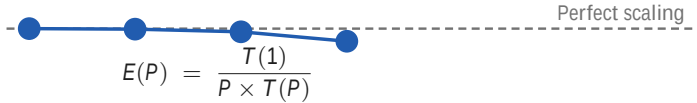

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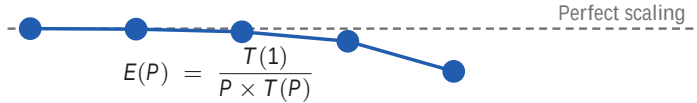

GPU-seconds of work, over GPU-seconds paid for

Workers	1	4	16	64	256
E□ciency, percent	**100**	**99.7**	**98.3**	**93.3**	**77.4**

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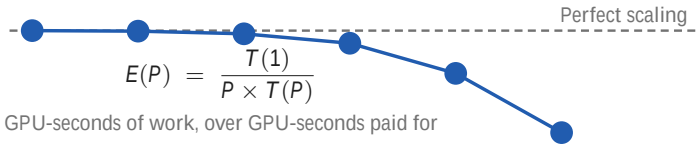
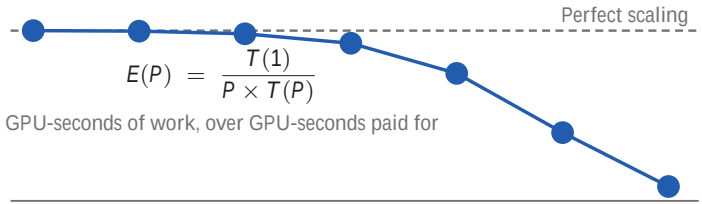
93 percent at 64 workers, 18 at 4,096



Workers	1	4	16	64	256	1,024
E□ciency, percent	**100**	**99.7**	**98.3**	**93.3**	**77.4**	**46.1**

Where it stops scaling

93 percent at 64 workers, 18 at 4,096



Workers	1	4	16	64	256	1,024	4,096
E□ciency, percent	**100**	**99.7**	**98.3**	**93.3**	**77.4**	**46.1**	**17.6**

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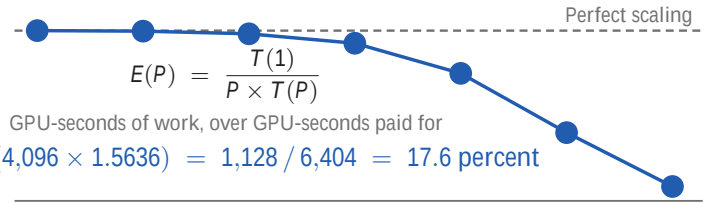
93 percent at 64 workers, 18 at 4,096



Workers	1	4	16	64	256	1,024	4,096
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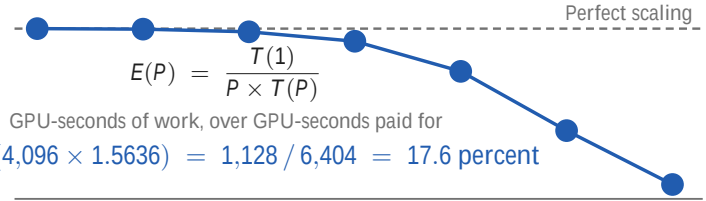

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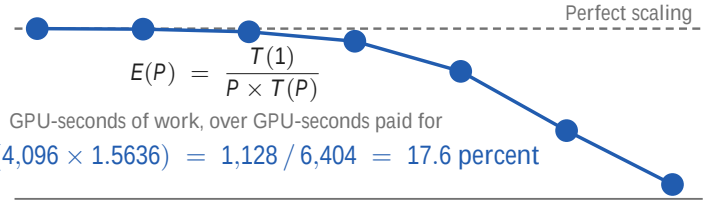

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- At 1,024 workers, communication is 54 percent of the step

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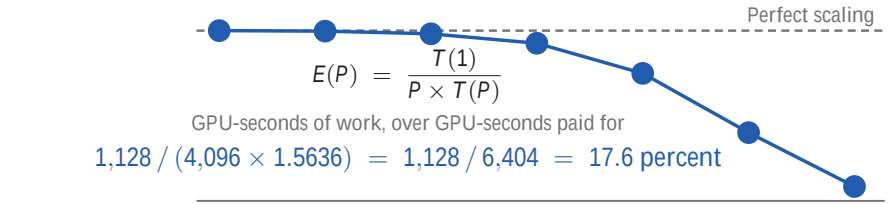

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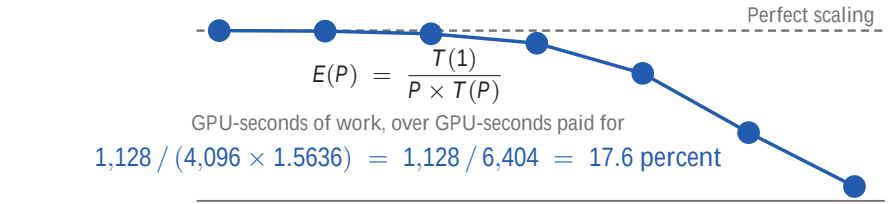

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- These are the model’s numbers, charging the all-reduce after the whole backward pass.

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In this model the step never falls below 1.29 seconds

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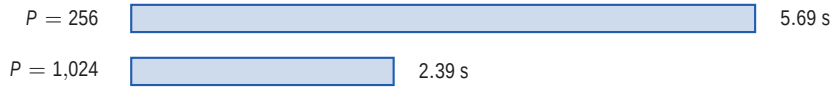

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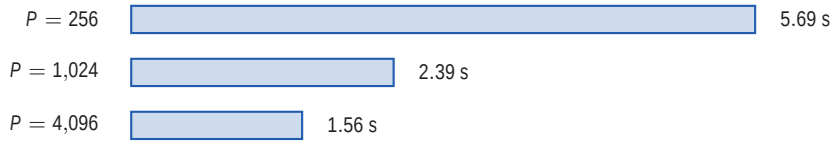

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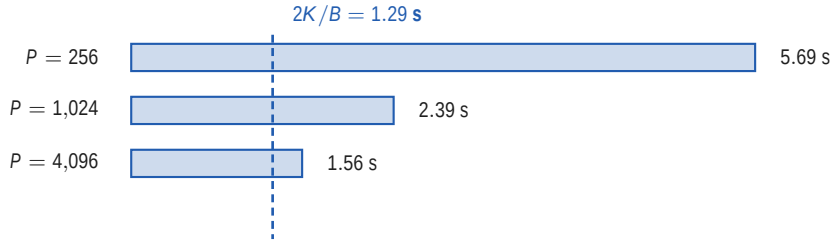

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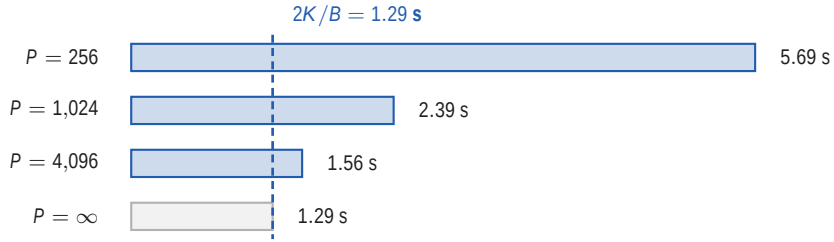

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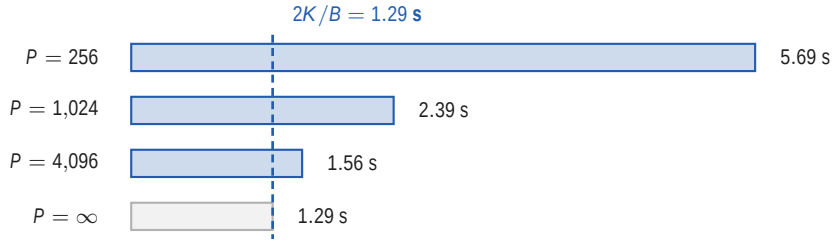

- Compute over P goes to zero, and $2(P-1)/P \times K/B$ goes to $2K/B$

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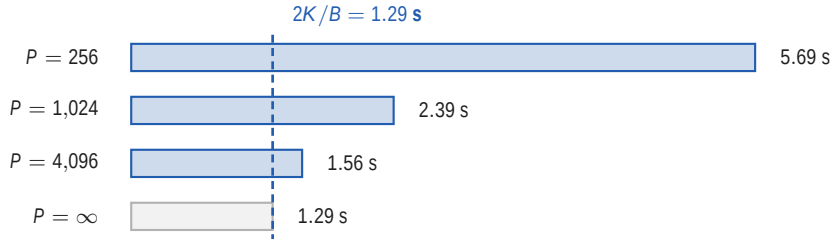

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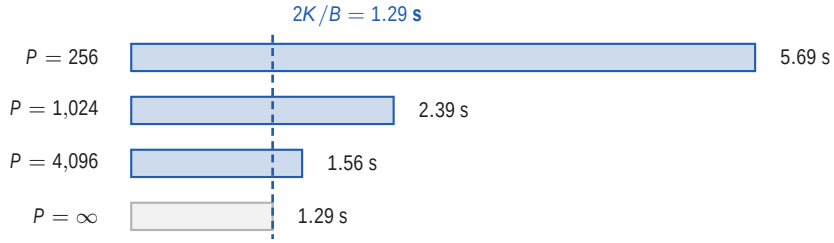

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- Four times the machine, 1,024 to 4,096, buys 0.83 s of a 2.39 s step
- Every worker after that, all of them together, buys 0.28 s more

Where it stops scaling

18 percent modelled, 80 percent measured

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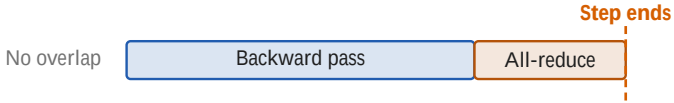

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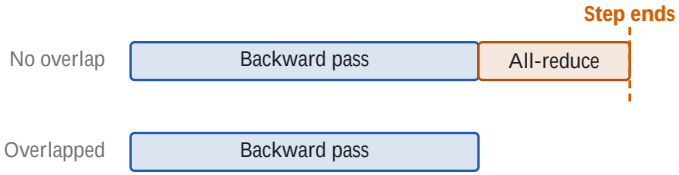

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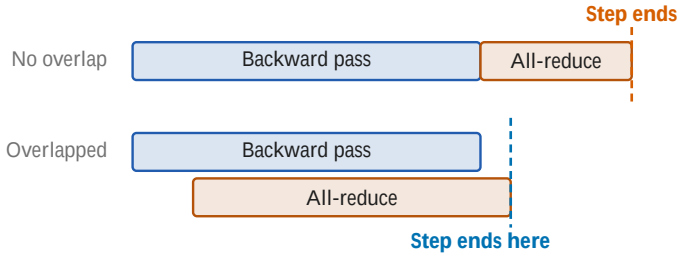

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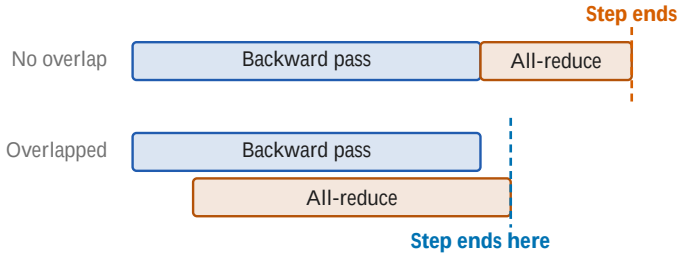

- The model above charges the all-reduce after the whole backward pass

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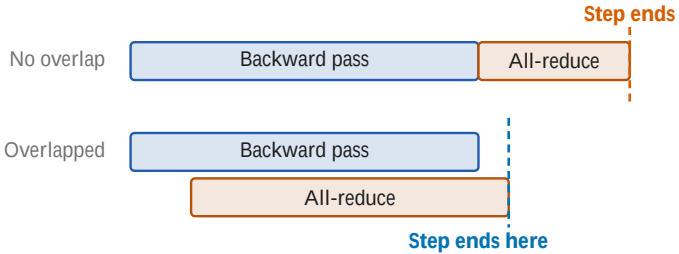

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- Real systems start averaging before the backward pass has finished

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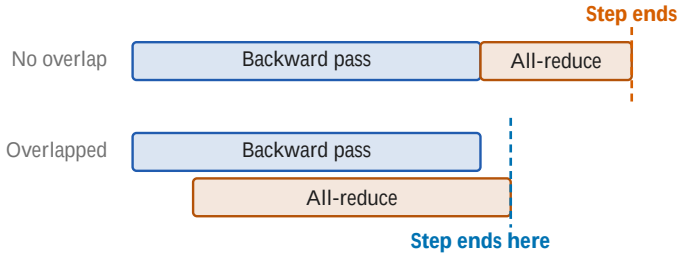
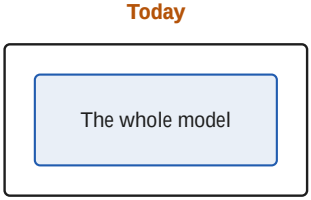


- The model above charges the all-reduce after the whole backward pass
- Real systems start averaging before the backward pass has finished
- Apertus measured about 80 percent at 4,096 GPUs, not 17.6, so overlapping is worth a factor of about 4.5

Where it stops scaling

Tomorrow the model does not fit



Where it stops scaling

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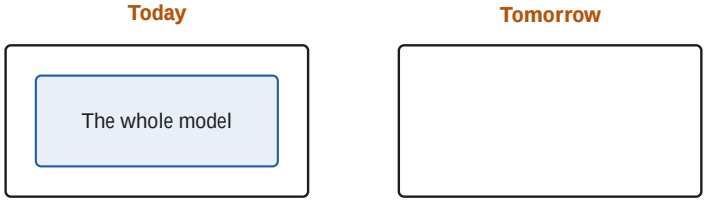

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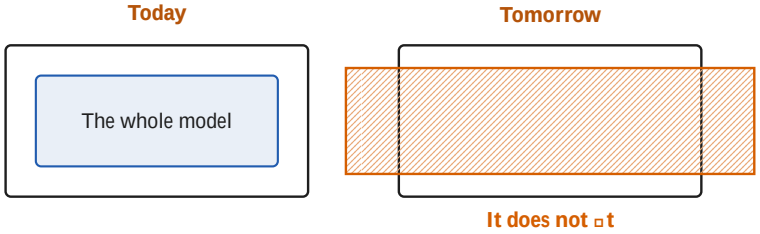

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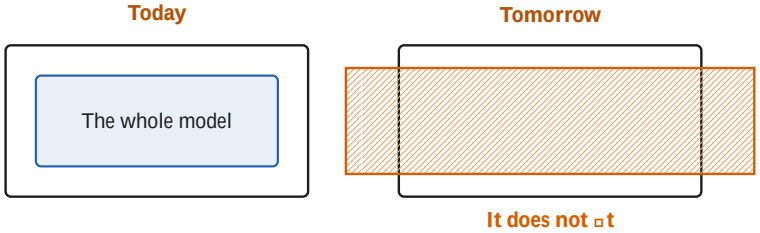
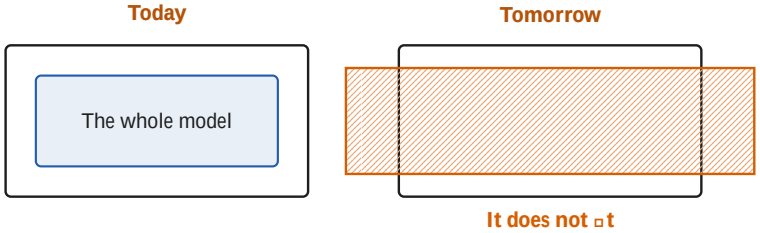


- Everything in this hour assumed one device holds the whole model

Where it stops scaling

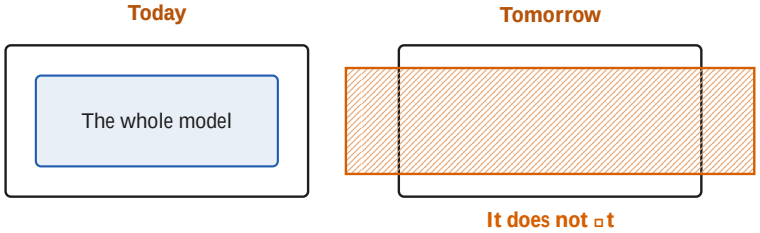
Tomorrow the model does not fit



- Everything in this hour assumed one device holds the whole model
- Apertus-70B does not

Where it stops scaling

Tomorrow the model does not fit



- Everything in this hour assumed one device holds the whole model
- Apertus-70B does not
- Tomorrow at 08:00: cut the model, not the batch